

Juan Roberto García Gómez

MES / ERP Systems Engineer · Full-Stack Developer · Industrial IoT

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68	~62%	~90%	6
machines, PLC stations and scanners connected — all in project scope	actual spend below the approved MES budget	less manual production reporting	plant areas on one data model

PROFILE

Manufacturing systems engineer specializing in MES, industrial integration and full-stack development. As IT Supervisor at a Korean automotive supplier (plastic injection molding and assembly), I designed and led the in-house development of the plant's production MES with ERP modules, from requirements to formal close. Hands-on with React, Node.js, Python, PostgreSQL and industrial protocols (EUROMAP 63, Modbus TCP, PLC); I lead a two-developer team and the plant's traceability pilot.

EXPERIENCE

IT Supervisor · Chunil México

Nov 2024 – Present

Korean automotive supplier · plastic injection molding and assembly · San Luis Potosí

Production MES with ERP modules — built in-house

- Designed and led the in-house development of the plant's MES (React, Node.js / Express, Python, PostgreSQL, REST APIs) to replace spreadsheet-based control; actual spend came in **~62% below the approved budget**, with vendors handling only legacy-PLC integration and cabling.
- Connected every machine in project scope — **34 injection molding machines** (ENGEL, WOOJIN, SUMITOMO) over EUROMAP 63 and Modbus TCP, **14 assembly stations** over PLC and **20 handheld scanners** — through Python microservices (collector, packager, reporting), putting **6 plant areas** on one data model and cutting manual production reporting by **~90%**.
- Built ERP modules for HR incidences, production planning, warehouse and purchasing; integrated BioTime (time clock) and CONTPAQi, which remains the accounting system, through a read-only SQL Server mirror. Added scheduled shift reports, Zebra ZPL labels and WhatsApp line-stop alerts.
- Engineered for shop-floor use: real-time dashboards (Socket.IO), JWT access control with 20 roles, 4 Expo / React Native scanner apps, **5,300+ automated tests** (unit and integration) and health monitoring with automatic recovery for 20 data services.

Project management

- Ran the MES as a formal project — requirements with Production, Quality, HR and Finance, vendor coordination and budget control — and **closed it in July 2026** with a final report to plant leadership and the CEO in Korea.
- Leading the **end-to-end traceability program** (resin to finished good), now in **pilot stage**: business case and BOM for **66 planned traceability points**, vendor comparison (ScanID, Higole, Xprinter, Zebra) against external integrator quotes, and **HQ approval (Sep 2026)** of a two-stage budget (pilot, then fleet). The pilot runs on the floor with touch kiosks, label printers and in-house software.

Team and plant IT

- **Hired and now lead two developers**, managing work in Jira and code in Git / GitHub.
- Administer plant IT — Microsoft 365, Cisco Meraki MX84 firewall, managed switches, servers, NAS backups and offboarding — and delivered a **network-security assessment and upgrade proposal** to leadership.

PROJECTS

OEE Live — Real-time OEE monitoring for injection plants without an MES

KeepCoding Full Stack final project · 2026

Polls 8 simulated injection machines every second, computes OEE per machine, shift and plant, and streams a live plant board with downtime classification, alerts and Excel export. 298 automated tests; containerized and deployed on Google Cloud.

Stack: Python (Modbus TCP collector) · FastAPI · Next.js · TypeScript · WebSocket · PostgreSQL · Redis · Docker · Google Cloud

oee-live-keepcoding.cloud.run Code · GitHub

PrecioRadar — Price comparison across 21 Mexican retailers in one search

Personal product · 2026

Searches 21 stores in parallel, matches the same product across differently named listings, recommends the best option and keeps a daily price history. Hexagonal API with 111 tests, a React web app and an Expo mobile app; scheduled jobs on Google Cloud.

Stack: TypeScript · Node.js · Fastify · Zod · React · Expo / React Native · PostgreSQL · Playwright · Docker · Cloud Run · Cloud SQL

precioradar-701205043940.us-central1.run.app

PREVIOUS EXPERIENCE

Claims, Loss & Recovery Coordinator — Oct 2023 – Nov 2024

Ramírez Valle & Asociados for General Motors San Luis Potosí · San Luis Potosí

- Recovered **MX$10.2M** in logistics claims within one year by structuring case analysis and evidence management.
- Managed **100+ investigations** across claims, supplier follow-up and executive reporting.
- Built Power BI dashboards that **cut analysis time by 40%** and gave management visibility of recovery status.

EDUCATION

Full Stack Web Development Bootcamp
KeepCoding · 2025 – 2026
Completed · final project 9.5/10

Business Analytics
Tecnológico de Monterrey · 2025

Big Data, AI & Machine Learning Bootcamp
KeepCoding · 2023 – 2024

B.A. in Economics
Universidad Autónoma de San Luis Potosí (UASLP) · 2018 – 2023
Degree awarded

B.Eng. in Industrial & Logistics Engineering
Universidad Interamericana para el Desarrollo (UNID) · 2020 – 2023
Degree awarded

LANGUAGES

Spanish	Native
English	Professional working proficiency

RECOGNITION

Department Employee of the Year 2025 — Chunil México

Live projects and portfolio
Scan to open the portfolio with the live demos

robertogag.github.io

CORE SKILLS

Manufacturing systems: MES · OEE · production planning · downtime tracking · lot traceability · quality / NG tracking · ERP integration · Zebra ZPL labels

Industrial IoT: EUROMAP 63 · Modbus TCP · PLC integration (Siemens LOGO!, LS Electric) · machine data collection · real-time dashboards (Socket.IO / WebSocket)

Software: JavaScript · TypeScript · Python · React · Next.js · Node.js · Express · Fastify · FastAPI · REST APIs · React Native (Expo)

Data: PostgreSQL · SQL Server · Redis · data modeling · ETL · Power BI

DevOps & quality: Git · GitHub Actions · Docker · Google Cloud (Cloud Run, Cloud SQL) · TDD · Vitest · pytest · Playwright

IT & management: Project management · business cases · budgeting · vendor evaluation · team leadership · Jira · Microsoft 365 · Cisco Meraki · network security